

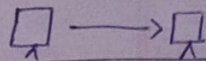
lec 2

Basic Of Internet

Date 17/09/24
Page _____

Evolution of Internet

Internet :- data transfer



not like sheet :- punching card

To store other than number like

'Hello'

or character

every letter has a ASCII value and then it is convert to binary

A - 65

a - 97

= \ ^ .

B - 66

b - 98

>] - ;

space also has an ASCII value - 32
(in dec)

[HELLO BHAIYA]

72 69

↓ ↓

ASCII value - 0 1 0 0 1 0 0 0

8 bit in binary

max

$2^8 = 256$ (characters)

How emoji, hindi, chinese characters stored?
ASCII was not sufficient.

- UTF - 8

- UTF - 16

- UTF - 32

$2^{16} = 65536$ characters

2^{32}

still not sufficient

↓
unicode
transformation
format

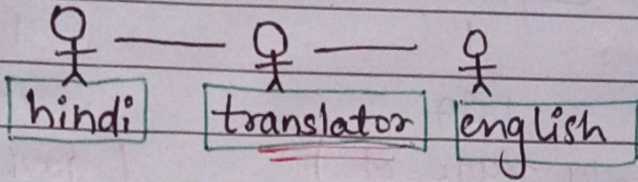
All the binary was written by humans
which was not feasible

→ Assembly language

□ - Assembler - □ ADD 2 3

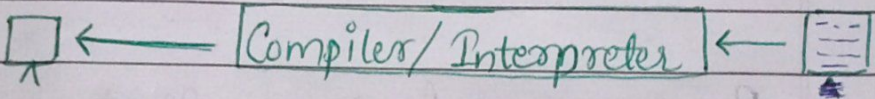
needs Translator

like



but we needed more optimised

→ High level language



Fast MLANG > Assembly > HLL

low level language C, C++

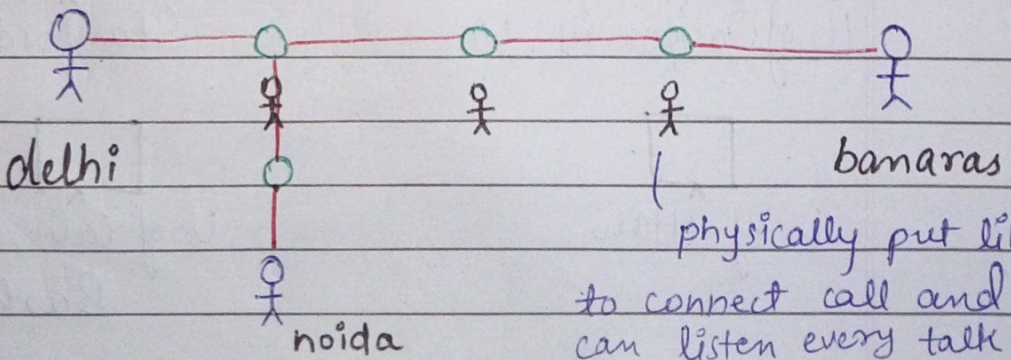
HLL → Java, python

ADD 2 3 → 2 + 3 (shift)

1900

Telephone

analog signal



→ Soviet Union v/s USA

cold war (not direct fight)

capitalism

1

USA

(freely trade of
goods & determine
prices by private
individuals) weapons/drugs

Yes

communism

1

Soviet Union

(govt. owns the
production, property
& other resources)

No

Russia reached space first
Then USA decided to reach moon.

→ Arpanet (1960) - by USA

To communicate between systems
& diff locations at times of even
nuclear war.



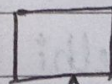
U. of California, LA



Stanford (SRI)



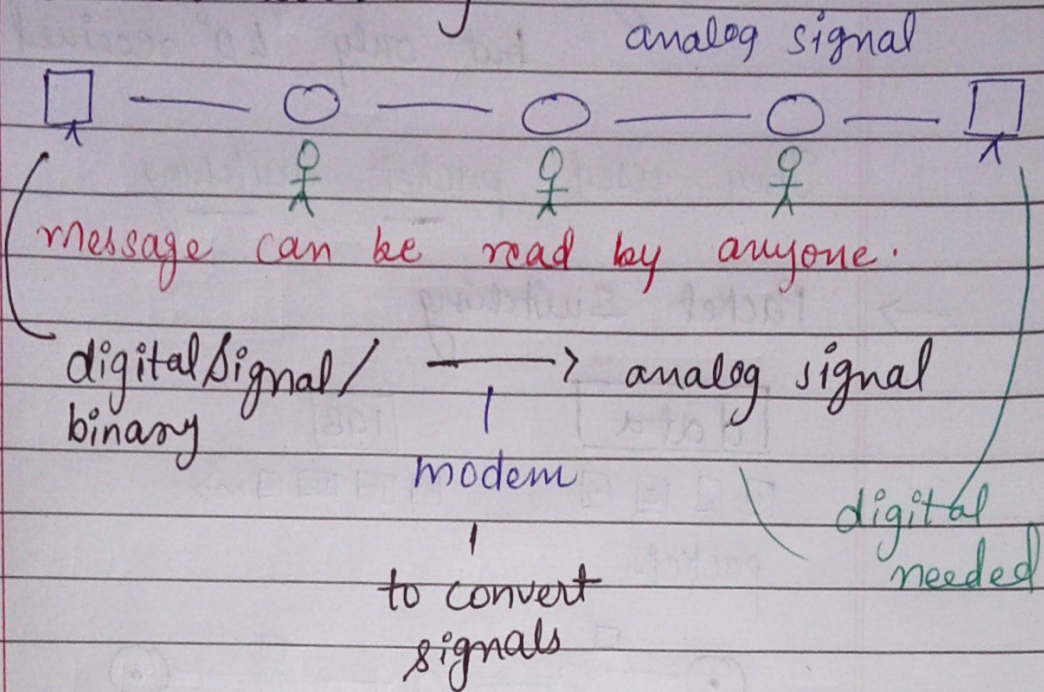
U. of Utah



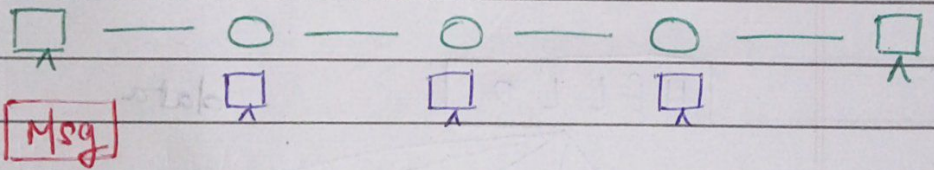
U. of Cali., Santa
Barbara

Before this,

→ Circuit Switching



→ Message Switching



No privacy issue but,
the msg will be whole wrap
up then only, transferred.
like whole YT video first loaded then play.

delay in network,
storage required

Then Arpanet came. (1960)

but got successful in 1969.

In 1969, they succeeded by sending a message "LOGIN".
but only "LO" received.

They used packet switching

→ Packet Switching

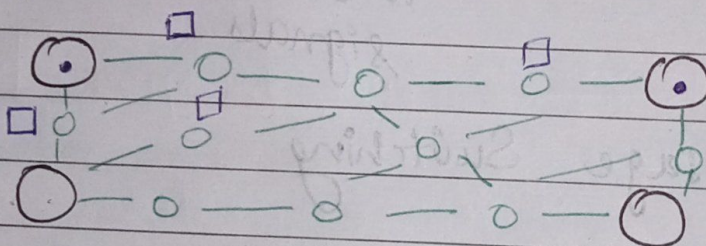
data

148

1 2 3 4

100 100 100 100 mb

packets



HELLO

data

H E L L O

packets